Task 44: Path Sum II

Problem Description

Given the root of a binary tree and an integer targetSum, return all root-to-leaf paths where the sum of the node values in the path equals targetSum. Each path should be returned as a list of the node values, not node references.

Java Solution

```
class Solution {
    public List<List<Integer>> pathSum(TreeNode root, int targetSum) {
        List<List<Integer>> res = new ArrayList<>();
        List<Integer> path = new ArrayList<>();
        dfs(root, targetSum, path, res);
        return res;
    }

    private void dfs(TreeNode root, int sum, List<Integer> path,
List<List<Integer>> res) {
        if (root == null) return;

        path.add(root.val);
        if (root.left == null && root.right == null && sum == root.val) {
            res.add(new ArrayList<>(path));
        } else {
            dfs(root.left, sum - root.val, path, res);
            dfs(root.right, sum - root.val, path, res);
        }
        path.remove(path.size() - 1);
    }
}
```